

Self-Supervised Monocular Depth Estimation From Videos via Adaptive Reconstruction Constraints

Xinchen Ye^{ID}, *Member, IEEE*, Yuxiang Ou, Biao Wu, Rui Xu^{ID}, *Member, IEEE*, and Haojie Li^{ID}

Abstract—To estimate depth maps from monocular videos in a self-supervised way, existing methods simultaneously predict the pose changes between adjacent frames and the depth maps of each frame, and then reconstruct the forward or backward frames using them, thereby casting depth estimation as a frame reconstruction problem. The corresponding reconstruction loss, which serves as a key supervision signal for training the whole network, can adversely affect the depth estimation accuracy if it is not properly established. In this paper, we propose a novel self-supervised monocular depth estimation method from videos via adaptive reconstruction constraints, i.e., designing the loss functions by establishing more accurate reconstruction constraints. Specifically, we first propose a pose-adaptive reconstruction loss to adaptively select the optimal pose parameterizations that yield the minimum reconstruction errors, reducing the impact of inaccurate posture on frame reconstruction. Then, we propose a region-sensitive reconstruction loss that fully utilizes the pretrained image reconstruction model to adaptively identify the poorly reconstructed regions and characterize the deviation of these regions on feature space. Finally, we additionally construct a multi-frame depth estimation network and design a reconstruction-guided bidirectional distillation loss to adaptively adjust the direction of distillation between networks of multi-frame and monocular depth estimation based on their current reconstruction quality, which encourages them to learn from each other and benefits the core task of monocular depth estimation. With our proposed losses, we achieve superior performance in comparison with state-of-the-art methods on benchmark datasets.

Index Terms—Unsupervised, depth estimation, monocular, adaptive reconstruction, video.

I. INTRODUCTION

MONOCULAR depth estimation is a fundamental and critical computer vision task that estimates the pixel-wise depth information from 2D images. Results from accurate depth estimation can facilitate various applications

such as virtual reality [1], scene reconstruction [2] and autonomous driving [3]. Despite the success of supervised methods [4], [5], [6], they are hindered by the need of time-consuming and labor-intensive ground truth annotation. As an alternative, existing video-based self-supervised methods [7], [8], [9] casts depth estimation as a frame reconstruction problem. They simultaneously predict the pose changes between two adjacent frames (called source frame and target frame respectively) and the depth map of target frame, and then reconstruct the source frame on the target view using the predicted pose and depth information. The photometric reconstruction loss, which measures the consistency between the reconstructed frame and the original target frame, serves as a key supervision for evaluating the quality of depth estimation and guiding the training of the whole network. However, when this loss is not effectively established, it can potentially lead to detrimental effects and hinder the accuracy of depth estimation.

Therefore, some current methods focus on improving the reconstruction loss from the following aspects: 1) utilizing feature-level reconstruction loss [10], [11] which leverages image features extracted from an additional pretrained image reconstruction auto-encoder model to strengthen the pixel fidelity in textureless areas; 2) identifying and excluding occluded areas using masking [12], [13], [14], or employing minimum reprojection of adjacent frames [7] to mitigate the photometric inconsistencies caused by occlusions; 3) developing regularization for brightness [15], [16], [17] to address illumination inconsistencies between adjacent frames. By modifying the reconstruction loss, the above methods can improve the accuracy of depth estimation to some extent, but many problems are still ignored when designing the reconstruction loss.

First, based on our observations of real-world driving scenarios, vehicles (or cameras) predominantly travel in straight trajectories, with turns occurring primarily at lane changes or intersections. This observation is further corroborated by an analysis of the KITTI dataset, as depicted in Figure 1, which highlights the imbalanced nature of rotation and translation in real-world scenarios, making it difficult for previous methods to simultaneously focus on the estimation of both rotation and translation in the pose network. It is worth noting that rotation is highly non-linear and often more difficult to estimate than translation. As a result, this limitation has a significant negative impact on the reconstruction loss, which in turn affects the performance of depth estimation.

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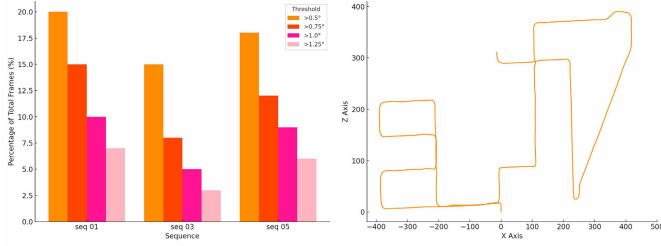


Fig. 1. Statistical results on the KITTI odometry dataset [19] consisting of 11 sequences. We use the ground truth poses to analyze the rotation between consecutive frames. To facilitate the interpretation, we convert the rotation matrices to Euler angles and examine the relative rotation angles to investigate the proportion of the rotations that exceed a certain angle threshold. The table specifically shows the percentages of rotation angles exceeding 0.5° , 0.75° , 1° , and 1.25° in sequences 01, 03, and 05 (left), and the overall trajectory plot for sequence 08 (right).

Second, although the feature-level reconstruction loss [11] can explicitly constrain each pixel to be discriminative in feature space, it does not fully exploit the capability of the pretrained image reconstruction auto-encoder model to emphasize on the feature inconsistency caused by poor frame reconstruction in self-supervised depth estimation. From the perspective of anomaly detection [18], the auto-encoder model pretrained on original target frames can obviously reconstruct a target frame better than an abnormal frame.¹ Thus, introducing such reconstruction deviations from the pretrained auto-encoder model into the loss design can better identify the poorly reconstructed regions in the abnormal frame and properly guide the training of the depth estimation network.

In addition, the monocular depth estimation network only acquires limited information because of the single-frame input, which fails to fully exploit temporal information from adjacent frames. Recent methods [20], [21], [22] have attempted to address this issue by constructing a teacher network to take more associative data (e.g., stereo images) as input to infer depth maps. The teacher provides more accurate depth information to effectively transfer knowledge to the monocular depth estimation network (student) through unidirectional distillation. However, these methods ignore the fact that the teacher also has errors in depth estimation and is even less accurate than student in some areas. Note that, monocular depth estimation relies mainly on the understanding of appearance and semantic knowledge within features, which can better handle occluded regions and avoid unreliable depth estimation issues that often arise in multi-view (temporal or stereo) depth estimation. Therefore, it is necessary to design mutual learning strategies to promote learning from respective strengths (*multi-view* vs. *monocular*) and ultimately facilitate the task of monocular depth estimation.

To address the aforementioned problems, our core idea is to design the loss functions by establishing more accurate *adaptive reconstruction constraints*, thus properly guiding the training of monocular depth estimation. As shown in Fig. 2, we first propose a pose-adaptive reconstruction loss to mitigate the impact of inaccurate pose estimation on frame

reconstruction. Instead of directly outputting a single 6-DoF pose vector, we propose to decompose the pose network into three parallel branches, each dedicated to independently estimating pure translation, pure rotation and full 6-DoF pose components. Upon this, we adaptively select the optimal pose parameterizations that yield the minimum reconstruction errors to establish the reconstruction loss, which can efficiently utilize the complementary information provided by different pose estimators. Second, we propose a region-sensitive reconstruction loss that fully utilizes the pretrained image reconstruction auto-encoder model to adaptively identify the poorly reconstructed regions on feature space. We first apply the warping and reconstruction selection on source frame to obtain a reconstructed frame, then form the loss by computing the difference between features of both reconstructed and target frames extracted from the pretrained model. Thus, the model can easily recognize the poorly reconstructed regions in the reconstructed frame as an anomaly, and provide significant deviated features than those of target frame, forcing the network to optimize these regions by back-propagating a greater loss caused by feature inconsistency. Finally, we additionally construct a multi-frame depth estimation network as a teacher and design a reconstruction-guided bidirectional distillation loss to adaptively adjust the direction of distillation between networks of multi-frame and monocular depth estimation based on their current reconstruction quality. This encourages them to learn from each other in an iterative collaborative training and benefits the core task of monocular depth estimation.

Through designing reasonable loss functions and imposing more accurate reconstruction constraints into the losses, our method achieves superior performance of self-supervised monocular depth estimation in comparison with state-of-the-art methods on benchmark datasets. Besides, our improvement focuses on the loss design and is suitable for assembly on any backbone network, which would not increase the runtime of depth estimation during testing. To summarize, our main contributions are listed as follows:

- We propose a pose-adaptive reconstruction loss to *adaptively* select the optimal pose parameterizations that yield the minimum reconstruction errors to establish the reconstruction loss, reducing the impact of inaccurate posture on frame reconstruction.
- We propose a region-sensitive reconstruction loss that fully utilizes the pretrained image reconstruction model to *adaptively* identify the poorly reconstructed regions as anomaly and characterize the feature deviation of these regions on feature space.
- We propose a reconstruction-guided bidirectional distillation loss to *adaptively* adjust the direction of distillation based on the reconstruction quality, encouraging both networks of multi-frame and monocular depth estimation to learn from each other.

II. RELATED WORKS

A. Self-Supervised Monocular Depth Estimation

We divide self-supervised depth estimation methods into two categories according to the supervisory signal. The first

¹In our scenario, an abnormal frame refer to the source frame synthesized on the target view using inaccurate estimation of pose and depth information.

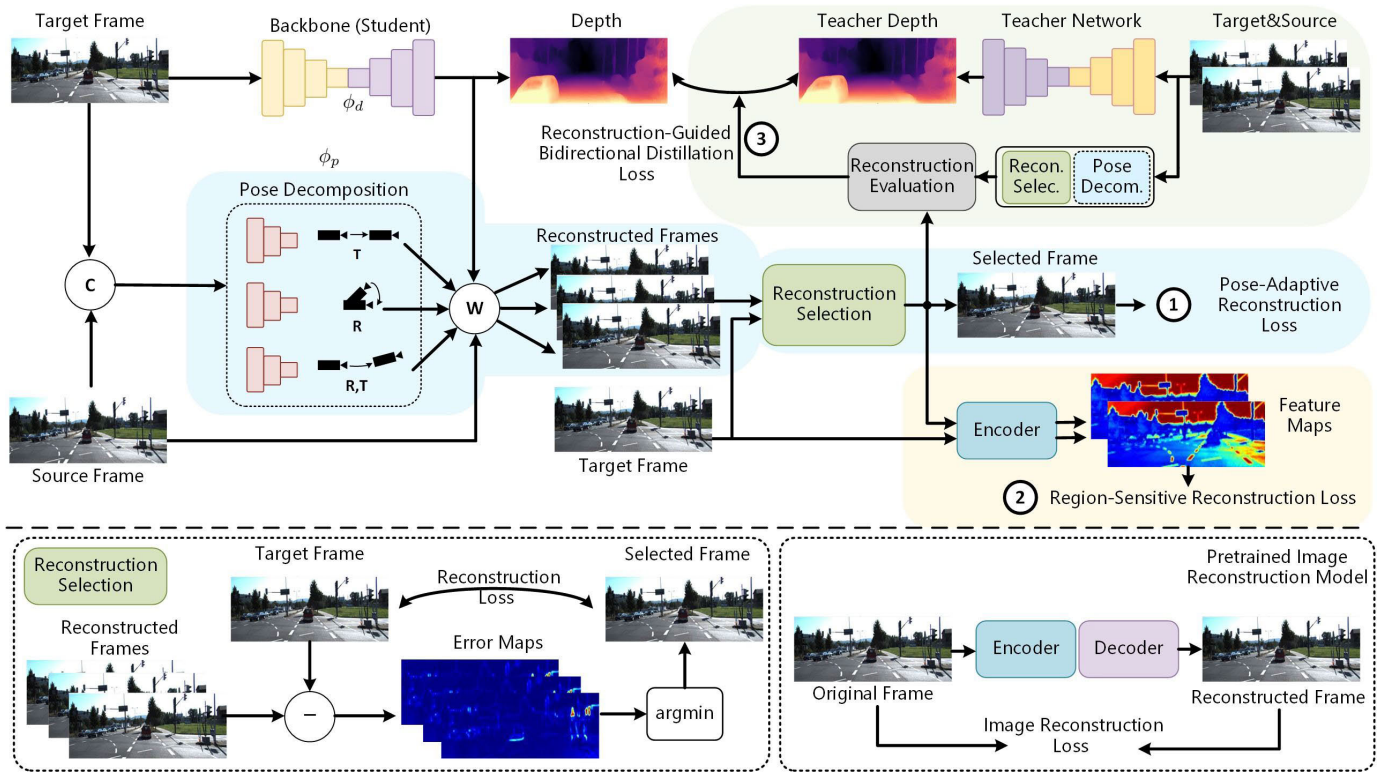


Fig. 2. Overview of our whole framework for self-supervised monocular depth estimation. Given a target frame and its adjacent source frame from a video, the goal is to predict a depth map from the target frame by a trainable backbone network. We highlight our contributions of three loss functions with different color rectangles. Through proper guidance of the proposed losses, the backbone network can be well trained.

category uses monocular sequences exclusively, while the second category uses additional auxiliary information such as stereo images, stereo videos, or semantics.

For the first category, SfM learner [23] first jointly learned monocular depth and ego-motion from monocular videos in a self-supervised way. However, such techniques still struggle with inferior depth estimation results when comparing with supervised methods because the photometric loss that plays a crucial role in the entire training process fails to establish effectively in cases such as occlusion and textureless areas. Numerous works are therefore suggested to address these limitations. Monodepth2 [7] leveraged masks to eliminate relatively static pixels and manages occlusion issues with minimal reprojection of adjacent frames. In order to establish effective supervision on textureless regions, feature-level reconstruction loss [10], [11] is applied to strengthen the pixel fidelity in textureless areas. Besides, some works also focus on the improvement of network architecture. For example, Bae et al. [24] proposed a CNN-Transformer hybrid network with multi-level feature aggregation to complement the shape bias and spatial locality bias to improve the performance and generalization of depth estimation.

For the second category, Garg et al. [25] first proposed to estimate stereo correspondences by minimizing a photometric loss between the left input image and the warped right image. Godard et al. [26] used consistency between the left and right views as a new constraint for training, reconstructing both the left and right views and calculating consistency to produce more accurate predictions. Ye et al. [20] utilized

recursive estimation and adaptive refinement to get more precise depth inference from stereo pairs, and then condensed the depth information into a monocular network. Further, many methods [7], [27], [28] employed stereo videos for training, which combines monocular sequence with stereo data as self-supervision signals. Besides, some works utilize semantic information to help depth estimation. Jung et al. [29] proposed a metric learning method that makes use of semantics-guided local geometry to optimize intermediate depth representations. Li et al. [30] proposed a semantic-aware spatial feature modulation scheme to relate depth distributions to the semantic category information.

In this paper, we focus on the task of video-based monocular depth estimation that uses only monocular sequences as supervision. Compared with stereo images/videos, monocular sequences are easier to obtain and apply in practical situations. However, methods based on monocular sequences need to bridge the relationship between adjacent frames by additionally estimating the camera ego-motion, which makes the task of depth estimation more difficult. Therefore, part of our work focuses on designing reasonable pose decomposition network and pose-adaptive reconstruction loss to mitigate the impact of inaccurate pose estimation on depth estimation.

B. Anomaly Detection

Anomaly detection establishes on the fundamental assumption that anomalous regions cannot be accurately reconstructed as they deviate from the learned normal patterns in training samples. These methods leverage generative models [31] to

establish pixel-level anomaly scores based on the pixel-wise reconstruction error. Alternative approaches [18] address anomaly detection as an inpainting problem, where patches from images are randomly masked and then predicted through the network. Thus, the pixel-wise reconstruction error can be computed as anomaly scores. Besides, embedding-based anomaly detection methods [32], [33] mainly project features into a compressed space and measure the distance between embedding vectors of normal training images and test examples. Among these methods, pre-trained deep neural networks are commonly used for feature extraction.

Inspired by the embedding-based anomaly detection, we propose a region-sensitive feature-level reconstruction loss that fully utilizes an image reconstruction model pretrained only on normal frames to identify the poorly reconstructed regions in abnormal frames and characterize the feature deviation of these regions on feature space.

C. Knowledge Distillation

Knowledge distillation, which is first proposed in [34] for condensing a large network into a smaller model, has garnered various interests recently [35]. Extensive works of monocular depth estimation also utilize the distillation technique to transfer knowledge from a sophisticated teacher network to a naive student network [20], [21], [36], [37]. Pilzer et al. [21] distilled the reconstruction difference between the original and re-synthesized input images to the student network through the computation of cycle inconsistency. Liu et al. [36] trained a stereo depth estimation network (teacher) to generate pseudo depth labels, then leverage the generated labels as supervision to distill more accurate depth information into the monocular network (student). Ye et al. [20] developed the recursive stereo distillation method to iteratively transfer depth information from a stereo network to a monocular network. Different from [20] and [36], Petrovai and Nedeveschi [37] constructed their teacher network by taking three consecutive frames as input instead of stereo images.

In general, teacher networks take more useful information as input (e.g., cycle inconsistency, stereo image, temporal frames) and generate more accurate results than those of students that takes only single image as input. Through distillation, performances of monocular depth estimation can be obviously improved. However, the above methods ignore the fact that the teacher also has errors in depth estimation. Upon this, we design a bidirectional distillation loss to promote mutual learning from respective strengths of both networks and ultimately facilitate the task of monocular depth estimation.

III. METHODS

A. Preliminaries

Given a target frame I_t and its adjacent source frame $I_s \in \{I_{t-1}, I_{t+1}\}$ from a video, the goal is to predict a depth map D_t from I_t by a trainable network ϕ_d . Let K be the known camera intrinsic matrix and $P_{t \rightarrow s}$ be the relative pose between I_t and I_s obtained by a trainable pose network ϕ_p , each point p_t in I_t can be warped into the corresponding pixel

p_s in the source view of I_s by:

$$p_s \sim K P_{t \rightarrow s} D_t(p_t) K^{-1} p_t \quad (1)$$

where $\sim$ represents the homogeneous equivalence. With Eq.(1), we can recover the target frame I_t from I_s by:

$$\hat{I}_t = I_s \langle \mathcal{W}(D_t, P_{t \rightarrow s}, K) \rangle, \quad (2)$$

where $\langle \cdot \rangle$ is a differentiable bilinear interpolation [38] and $\mathcal{W}(\cdot, \cdot, \cdot)$ is the warping operation corresponding to Eq. (1).

Following [7], the training loss is the photometric reconstruction errors between the target frame I_t and the reconstructed frame $\hat{I}_t$, based on the $L1$ and SSIM metrics, which is defined as:

$$L_p(\hat{I}_t, I_t) = \alpha \frac{1 - \text{SSIM}(\hat{I}_t, I_t)}{2} + (1 - \alpha) |\hat{I}_t - I_t|, \quad (3)$$

where α is set to 0.85. Also, a smoothness loss [7] is added to encourage the predicted depth map to be smooth, which is computed as follows:

$$L_s = |\partial_x D_t| e^{-|\partial_x I_t|} + |\partial_y D_t| e^{-|\partial_y I_t|} \quad (4)$$

Besides, we employ the minimum photometric error, auto-masking and multi-scale depth loss techniques introduced in [7] to reduce the negative impact of occlusion and stationary objects, thus further improving the performance of baseline.

B. Network Architecture

Our framework is based on the above self-supervised paradigm which contains a backbone monocular depth estimation network ϕ_d and a pose network ϕ_p . Upon this, we novelly design three loss functions by establishing adaptive reconstruction constraints, thus properly guiding the training of whole network. Specifically, as shown in Fig.2, we first propose a pose-adaptive reconstruction loss to mitigate the impact of inaccurate pose estimation on frame reconstruction. Instead of directly outputting a single 6-DoF pose vector, we introduce a pose decomposition network to independently estimate pure translation and pure rotation, together with the 6-DoF pose vector. Through warping, we obtain a set of reconstructed frames based on different poses and adaptively select an optimal one corresponding to the minimum reconstruction errors from the reconstructed frames through the reconstruction selection module. Upon this, the photometric reconstruction loss L_p can be established by computing the difference between the target frame and the selected frame.

Second, we propose a region-sensitive reconstruction loss that fully utilizes the pretrained encoder of an image reconstruction model to adaptively identify the poorly reconstructed regions on feature space. We send the selected frame and the target frame into the encoder to extract their respective feature maps and form the loss by computing the difference between them. Note that, the encoder pretrained on original frames can provide significant deviated features of poorly reconstructed regions (regarded as anomaly) in reconstructed frames compared to those of target (original) frames, forcing the network to optimize these regions by back-propagating a greater loss caused by feature inconsistency.

TABLE I
LIST OF SYMBOLS AND NOTATIONS

Symbol	Description
ϕ_d	Monocular depth estimation network
ϕ_p	Pose network
$\hat{I}_t^i$	Set of reconstructed frames
L_p^i	Photometric reconstruction loss for each $\hat{I}_t^i$
i^*	Optimal pose parameterization
$\hat{I}_t^{i^*}$	Selected reconstruction frame
$L_p^{i^*}$	Corresponding reconstruction loss for i^*
τ	Mask for determining distillation direction
stu	Superscript related to the student network
tea	Superscript related to the teacher network

Finally, we additionally construct a teacher depth estimation network with adjacent multiple frames as input, which enables the network to better model the temporal information. Upon this, we design a reconstruction-guided bidirectional distillation loss to adaptively adjust the direction of distillation between the student and teacher networks. Once obtaining the selected frames from reconstruction selection of both networks, we compute the distillation direction mask by comparing their reconstruction qualities through the reconstruction evaluation module and adaptively steer the distillation direction, which can encourage both networks to learn from each other in an iterative collaborative training and benefits the student. The symbols and notations used are summarized in Table I.

C. Pose Decomposition & Pose-Adaptive Reconstruction Loss

To accurately predict the pose transformation, we design a pose decomposition network, which contains three parallel branches to independently predict pure translation $T_{t \rightarrow s} \in \mathbb{T}(3)$, pure rotation $R_{t \rightarrow s} \in \mathbb{SO}(3)$ and full 6-DoF pose $P_{t \rightarrow s} \in \mathbb{SE}(3)$. According to Eq.(1) and Eq.(2), we generate a set of reconstructed frames $\{\hat{I}_t^i, i \in (\mathbb{T}(3), \mathbb{SO}(3), \mathbb{SE}(3), \mathbb{T}(3) + \mathbb{SO}(3))\}$ by considering four pose transformation cases, i.e., $T_{t \rightarrow s}$, $R_{t \rightarrow s}$, $P_{t \rightarrow s}$ and the combined pose $T_{t \rightarrow s} + R_{t \rightarrow s}$.

Then, in the reconstruction selection module, we subtract $\hat{I}_t^i$ from the target frame I_t and generate reconstruction errors for each $\hat{I}_t^i$ through the photometric reconstruction loss $L_p^i(\hat{I}_t^i, I_t)$ shown in Eq.(3). By comparing these error values and selecting the one with the minimum reconstruction error, we obtain the optimal pose parameterizations:

$$i^* = \arg \min_i L_p^i \quad (5)$$

Upon this, we can finally select the reconstruction frame $\hat{I}_t^{i^*}$ and its corresponding reconstruction loss $L_p^{i^*}$ to form our pose-adaptive reconstruction loss. During back propagation for current iteration of training, we only update the i^* -th branch to make itself to be more specialized in estimating its own pose transformation. When the combined pose $T_{t \rightarrow s} + R_{t \rightarrow s}$ is chosen to be optimal, both translation and rotation branches will be updated. Through our improvement, the complementary information provided by different pose estimators can be efficiently utilized to mitigate the impact of inaccurate pose estimation on reconstruction loss.

D. Region-Sensitive Reconstruction Loss

In [11], the authors proposed to utilize feature-level reconstruction loss to strengthen the pixel fidelity and discriminant in textureless areas on feature space. However, they just leverage the pretrained image reconstruction model (shown in Fig. 2) as a feature extractor, but ignore the potential capabilities of the model to identify the poorly reconstructed frames as anomaly, which motivate us to go a step further.

Unlike [11] that directly extracts features of original source and target frames from the pretrained model and then applies warping operation to align source features to the target view for the convenience of per-pixel loss computation, we invert the entire process. That is, we first apply the pipeline of warping and reconstruction selection on source frame and then extract features of both the selected frame $\hat{I}_t^{i^*}$ and the target frames I_t from the pretrained model. Since the model is trained on normal original frames, it cannot easily reconstruct the abnormal regions (caused by wrong depth information in the warping) in $\hat{I}_t^{i^*}$ and provide significant deviated features than those of target frame I_t . Upon this, we obtain our region-sensitive reconstruction loss, which is defined as follows:

$$L_r = \|\phi_e(\hat{I}_t^{i^*}) - \phi_e(I_t)\|_1 \quad (6)$$

where ϕ_e is the pretrained encoder of an image reconstruction model that is obtained similar to the way introduced in [11].

E. Reconstruction-Guided Bidirectional Distillation Loss

The teacher network has the same architecture with the student network except for its input that contains the target frame and two adjacent source frames $\{I_{t-1}, I_t, I_{t+1}\}$, which can better capture the geometric structure of the scene and help train the student. However, it is important to note that in scenarios with occlusions and moving objects, depth estimation results from the teacher are often unreliable. Therefore, the key idea to promote collaborative learning between both networks is to ensure that only relatively accurate knowledge is distilled from the teacher to the student, and vice versa, which motivate us to design the bidirectional distillation loss.

Specifically, assume that we frozen the teacher network and train the student network in the current training stage, we can obtain respective reconstructed frames $\hat{I}_t^{tea}, \hat{I}_t^{stu}$ and their corresponding photometric reconstruction errors L_p^{tea}, L_p^{stu} from the reconstruction selection module.² According to the pixel-wise comparison between L_p^{tea} and L_p^{stu} , a mask τ is generated to determine whether the depth information of each pixel should be distilled from the teacher to the student:

$$\tau(k) = [L_p^{stu}(k) > L_p^{tea}(k)] \quad (7)$$

Thus, the distillation loss can be defined as:

$$L_d = \|\tau \circ (D_t - D_t^{tea})\|_1 \quad (8)$$

Here, D_t^{tea} represents the estimated depth map by the teacher and ‘ $\circ$ ’ denotes element-wise multiplication. It indicates that we only use the depth pixels of the teacher to supervise the

²The superscript of i^* is omitted for easy presentation.

student when their corresponding reconstruction errors are smaller than those of student.

After the student is trained, we frozen it and train the teacher in the next training stage by an iterative fashion with the same distillation strategy introduced above, which can be defined as:

$$L_d^{tea} = \|(1 - \tau) \circ (D_t - D_t^{tea})\|_1 \quad (9)$$

F. Total Loss

The training process mainly consists of three stages. At beginning, we train the student network in a fully self-supervised manner and the loss function incorporates three components, i.e., pose-adaptive reconstruction loss L_p^* , region-sensitive reconstruction loss L_f and the smoothness loss L_s , which can be written as:

$$L_{stu} = \lambda_s L_s + \lambda_f L_f + \mu L_p^* \quad (10)$$

where λ_s and λ_f are the weighting factors that control the importance of each loss. μ refers to the mask proposed in [7]. Similarly, the teacher network is trained from scratch with the same loss of L_{stu} .

Then, we fine-tune the teacher with the help of student. The depth map of teacher network is adaptively supervised by that of the student. As shown in Eq.(7) and Eq.(9), we compute the mask τ and distill the knowledge from student to teacher with L_d^{tea} . For the reconstruction losses, we also introduce the mask τ and revise the losses to avoid conflicts with the distillation loss:

$$L_p^{tea} = (1 - \tau) \cdot L_p^{stu} + \tau \cdot L_p^{tea} \quad (11)$$

$$L_f^{tea} = (1 - \tau) \cdot L_f^{stu} + \tau \cdot L_f^{tea} \quad (12)$$

where L_p^{stu} , L_f^{stu} represent the reconstruction loss of the student network while L_p^{tea} , L_f^{tea} are for the teacher. Consequently, the loss for training the teacher network is:

$$L_{tea} = \lambda_s L_s + \lambda_f L_f^{tea} + \mu L_p^{tea} + \lambda_d L_d^{tea} \quad (13)$$

where μ , λ_s and λ_f are hyper-parameters similar to those used in L_{stu} . λ_d is the weighting factors of distillation.

Finally, we freeze the teacher and use it to guide the training of the student. The loss function in this stage is similar to the loss function L_{tea} used for training the teacher network, and we omit it for saving the space. Note that, the above training pipeline can be iteratively conducted between the teacher and student until the convergence.

IV. EXPERIMENTS

Training Dataset: We evaluate the proposed method on KITTI [19] and Make3D [49] datasets. KITTI dataset [19] comprises monocular sequences and corresponding 3D laser scans of outdoor scenes captured by imaging equipment mounted on a moving vehicle. For training purposes, we adopt the dataset split proposed by [39]. After eliminating static frames using a pre-processing step recommended by [23], we are left with 39,810 monocular frame triplets for training and 4,424 frame triplets for testing. Make3D [49] consists of 400 images as training set and 134 images as testing set. Since Make3D only includes RGB-D pairs without adjacent

frames, it cannot be used for training, and thus we only test the generalization on Make3D through the well-trained KITTI model. Besides, we also test the adaptability of our method on NYU v2 dataset [50], a rotation-dominant indoor dataset. In NYU v2, we use the officially provided 654 densely labeled images for testing, and the rest sequences for training.

Implementation Details: The framework mainly contains four parts, i.e., student and teacher networks for depth estimation, pose network in pose decomposition, and the encoder pretrained from an image reconstruction model. For depth estimation, we use the network architecture that replaced the original ResNet-based depth encoder in Monodepth2 [7] with HRNet-18 [51] as our backbone for both student and teacher, but adjust the number of channels for multiple frame inputs through an 1×1 convolution to adapt to the teacher network. For pose estimation, we use the generic pose network proposed in [7] as our backbone, which employs ResNet-18 [52] as the encoder. For the pretrained encoder, we employ the same method in [11] to construct an image reconstruction auto-encoder model, and use its encoder as a frozen module to extract features for computing the feature-level reconstruction loss.

During the training process, we initially train the student network and teacher network from scratch until they converge. Subsequently, we fine-tune the teacher network through selective distillation, utilizing the fixed student network as a reference. Then, we leverage the more capable teacher network to assist the training of student network by fixing the teacher and training the student until it converges again. The number of iterations in collaborative training is set to 2 according to our experimental verification. We set the batch size as 10 and adopt the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-4}$. We set the initial learning rate as 10^{-4} for the first 15 epochs and then 10^{-5} for fine-tuning the remainder. For the adjustment parameters, λ_s , λ_f , and λ_d are respectively set to 10^{-3} , 10^{-3} , and 1 according to experimental verification. The whole training process is shown in Algorithm 1. We adopt the measures used in [39] for quantitative evaluation.

A. Performance Comparison

1) *Quantitative Result:* Table II shows the comparison results with state-of-the-art methods. Note that, all the compared methods use monocular images as input in testing phase. For fair comparison, we only compare with methods that exclusively using monocular videos (M) as supervision, and without any additional data incorporated. As a result, our method exceeds other methods including the transformer-based ones (i.e., MonoFormer and Lite-Mono-8M), especially on the Sq Rel indicator, where it attains a 5% increase over the runner-up, proving its effectiveness.

2) *Qualitative Result:* Fig. 3 shows the qualitative results of our depth predictions. Compared to other self-supervised methods, our method can recover the details of thin objects and the fine structures of objects (highlighted by the red boxes in the figure), and achieve the lowest RMSE values on each given example. For instance, in the first column, our method can recover the depth of the billboards and fence posts,

TABLE II

RESULTS ON KITTI DATASET USING THE EIGEN SPLIT [39]. ALL METHODS ARE TRAINED BY USING MONOCULAR VIDEOS WITH NO ADDITIONAL DATA INCORPORATED. THE BEST RESULTS ARE MARKED WITH BOLD-FACE WHILE SECOND BEST ONES WITH UNDERLINE RESPECTIVELY

Method	Year	Error Metric (lower is better)				Accuracy Metric (higher is better)		
		Sq Rel	RMSE	RMSE log	Abs Rel	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
HR-Net [40]	ICCV'19	0.837	4.945	0.197	0.121	0.853	0.955	0.982
Monodepth2 [7]	ICCV'19	0.903	4.863	0.193	0.115	0.877	0.959	0.981
PackNet [41]	CVPR'20	0.785	4.601	0.189	0.111	0.878	0.960	0.982
FeatDepth [11]	ECCV'20	0.729	4.481	0.179	0.104	0.893	<u>0.965</u>	0.984
HR-Depth [42]	AAAI'21	0.792	4.632	0.185	0.109	0.884	0.962	<u>0.983</u>
CADepth [43]	3DV'21	0.769	4.535	0.181	0.105	0.892	0.964	<u>0.983</u>
DIFFNet [28]	BMVC'21	0.764	4.483	0.180	0.102	<u>0.896</u>	<u>0.965</u>	<u>0.983</u>
RM-Depth [44]	CVPR'22	<u>0.710</u>	4.513	0.183	0.108	0.884	0.964	<u>0.983</u>
BRNet [45]	ECCV'22	0.698	4.462	0.179	0.105	0.890	<u>0.965</u>	0.984
MonoFormer [24]	AAAI'23	0.846	4.580	0.183	0.104	0.891	0.962	0.982
Lite-Mono-8M [46]	CVPR'23	0.729	<u>4.454</u>	<u>0.178</u>	<u>0.101</u>	0.897	<u>0.965</u>	<u>0.983</u>
SC-DepthV3 [47]	TPAMI'23	0.756	4.709	0.188	0.118	0.864	0.960	0.984
Xiong et al. [48]	EAAI'24	0.868	4.986	0.200	0.122	0.857	0.953	0.980
Ours	/	0.675	4.386	0.176	0.100	0.893	0.966	0.984

Algorithm 1 Depth Estimation Process

Input: Monocular video frames $\{I_t\}$

Output: Estimated depth maps $\{D_t\}$

Model Training:

Phase 1: Student Training

for each frame I_t in each training epoch **do**

1. Estimate depth map D_t^{stu} using ϕ_d^{stu}
2. Estimate poses $T_{t \rightarrow s}$, $R_{t \rightarrow s}$, $P_{t \rightarrow s}$ using ϕ_p
3. Generate reconstructed frames $\{\hat{I}_t^i\}$ for different poses

transformations

4. Compute photometric reconstruction losses $\{L_p^i\}$ for

each $\hat{I}_t^i$

5. Select optimal pose $i^* = \arg \min_i L_p^i$
6. Compute pose-adaptive reconstruction loss $L_p^{i^*}$ using

$\hat{I}_t^{i^*}$

7. Compute region-sensitive reconstruction loss L_r

8. Update ϕ_d^{stu} using total loss L_{stu}

end for

Phase 2: Teacher Training

Freeze student network ϕ_d^{stu}

for each frame I_t in each training epoch **do**

1. Estimate depth map D_t^{tea} using ϕ_d^{tea}
2. Compute photometric reconstruction errors L_p^{stu} and L_p^{tea} through pose-adaptive reconstruction loss

3. Generate mask τ by comparing L_p^{stu} and L_p^{tea}

4. Compute distillation loss L_d^{tea} for teacher network:

L_d^{tea}

5. Compute region-sensitive reconstruction loss L_r

6. Update ϕ_d^{tea} using total loss L_{tea}

end for

Phase 3: Iterative Updating

Swap roles of student and teacher networks and repeat Phase 2

Model Testing:

Use the trained student network ϕ_d^{stu} to estimate depth maps D_t for test frames I_t

to recognize the wall on the left side of image. In the sixth column, it is clear that our method provide the most accurate depth prediction, both in terms of object perception and RMSE numerical results.

B. Ablation Study

1) *The Full Pipeline:* In this section, we analyze the performance of our method by studying the influence of three losses we have proposed on the final depth prediction. We compared the performances with and without our proposed losses and reported the results in Table IV. At the starting point, the first row gives our baseline that replaced the original ResNet-based depth encoder in Monodepth2 [7] with HRNet-18 [51], and the same self-supervision pipeline introduced in the Sec. III-A. Upon this, we replace the pose network [7] with our pose decomposition module and corresponding pose-adaptive reconstruction loss (the second row), and the performance of depth estimation is obviously improved, especially on Sq Rel and RMSE metrics (15.5% and 4.8% improvement respectively). Similarly, introducing the region-sensitive reconstruction loss solely (the third row) can also bring an obvious improvement against the baseline. Going a step further, introducing both of them into the baseline (the fourth row) can push the performance to a new upper bound. Finally, when further using the bidirectional distillation to encourage a mutual learning from both teacher and student networks (the last row), the ultimate superior performance is achieved. Through the ablation study, each proposed loss can present its contribution to the performance of depth estimation.

Next, we give some detailed analysis through visual and numerical results to further demonstrate the effectiveness of our proposed losses:

2) *Pose-Adaptive Reconstruction Loss:* Fig. 4 shows the error maps of reconstructed images against groundtruth target images without and with pose decomposition module and pose-adaptive reconstruction loss. It is obvious that, under the condition that the depth estimation network remains unchanged, training with the loss will make the reconstruction error smaller (highlighted by the red rectangles). Since the displacement of the foreground between two frames will be

while other methods predict inaccurate depths or wrong shape. In third column, Monodepth2, HR-Depth, and Lite-Mono fail



Fig. 3. Qualitative comparison with other methods on KITTI dataset: (a) Monodepth2 [7], (b) HR-Depth [42], (c) DIFFNet [28], (d) Lite-Mono-8M [46], (e) Ours. Regions of red rectangles in predicted depth maps present obvious visual differences among these methods. Besides, error maps (red for higher errors while blue for lower errors) against groundtruths and their corresponding RMSE values are given for better presentation.

TABLE III

ABLATION STUDY OF THE PROPOSED POSE-ADAPTIVE RECONSTRUCTION LOSS (PAR), REGION-SENSITIVE RECONSTRUCTION LOSS (RSR), AND RECONSTRUCTION-GUIDED BIDIRECTIONAL DISTILLATION LOSS (RGBD) ON KITTI DATASET

PAR	RSR	RGBD	Error Metric (lower is better)				Accuracy Metric (higher is better)		
			Sq Rel	RMSE	RMSE log	Abs Rel	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
			0.799	4.609	0.186	0.108	0.888	0.963	0.982
✓			0.730	4.442	0.182	0.102	0.894	0.964	0.983
	✓		0.747	4.498	0.183	0.103	0.893	0.963	0.982
✓	✓		0.720	4.430	0.182	0.101	0.896	0.965	0.983
✓	✓	✓	0.675	4.386	0.176	0.100	0.893	0.966	0.984

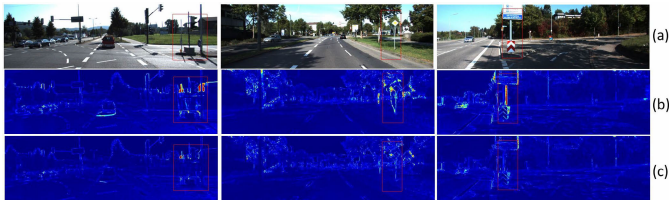


Fig. 4. Qualitative ablation study of the proposed pose-adaptive reconstruction loss. Error maps (red for higher errors while blue for lower errors) of reconstructed images (b) without and (c) with pose-adaptive reconstruction loss against groundtruth target images are shown.

TABLE IV

QUANTITATIVE RESULT OF THE POSE ESTIMATION RESULT ON KITTI ODOMETRY DATASET

Method	Seq. 09			Seq. 10		
	RTE	RRE	ATE	RTE	RRE	ATE
SfMLearner [23]	19.15	6.82	0.016	40.40	17.69	0.013
MatchVO [53]	9.91	3.8	0.009	12.18	5.9	0.008
MonoDepth2 [7]	17.17	3.85	-	11.68	5.31	-
SC-Depth [54]	7.31	3.05	-	7.79	4.90	-
Baseline	5.67	1.47	0.020	6.11	1.66	0.014
Baseline + PAR	5.07	1.42	0.007	5.84	1.62	0.005

relatively large, inaccurate pose estimation will cause wrong re-projection, resulting in larger reconstruction error, especially around the fine structures of foreground objects. This also proves that our pose estimation is more accurate and avoids interference to the depth estimation network, which indirectly helps improve the performance of depth estimation.

Moreover, we test the effectiveness of our pose decomposition network on the KITTI odometry dataset against the

baseline (without our proposed pose modules). As shown in Table IV, absolute trajectory error (ATE), relative translation error (RTE), and relative rotation error (RRE) are given. Please note that ATE is calculated following the approach in SfMLearner [23], while RTE and RRE are computed based on SC-Depth [54]. RTE is the average translational root mean square error (RMSE) drift in percentage on length from 100, 200, ..., 800m, while RRE is the average rotation RMSE

TABLE V
ABLATION STUDY OF THE PROPOSED REGION-SENSITIVE RECONSTRUCTION LOSS AGAINST THE FEATURE METRIC LOSS [11]

Method	Error Metric (lower is better)				Accuracy Metric (higher is better)		
	Sq Rel	RMSE	RMSE log	Abs Rel	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
Baseline	0.799	4.609	0.186	0.108	0.888	0.963	0.982
Baseline + [11]	0.768	4.508	0.184	0.105	0.890	0.963	0.982
Baseline + RSR (Ours)	0.747	4.498	0.183	0.103	0.893	0.963	0.982

TABLE VI
ABLATION STUDY OF THE PROPOSED BIDIRECTIONAL DISTILLATION LOSS. ‘TEACHER_INITIAL’ AND ‘STUDENT_INITIAL’ REPRESENT THE CONVERGED RESULTS WHEN TRAINING FROM SCRATCH WITHOUT BIDIRECTIONAL DISTILLATION, WHILE ‘TEACHER_END’ AND ‘STUDENT_END’ ARE THE CONVERGED RESULTS WITH DISTILLATION

Method	Error Metric (lower is better)				Accuracy Metric (higher is better)		
	Sq Rel	RMSE	RMSE log	Abs Rel	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
Teacher_Initial	0.702	4.415	0.180	0.100	0.897	0.965	0.983
Student_Initial	0.720	4.430	0.182	0.101	0.896	0.965	0.983
Teacher_End	0.690	4.405	0.179	0.099	0.899	0.966	0.983
Student_End	0.675	4.386	0.176	0.100	0.893	0.966	0.984

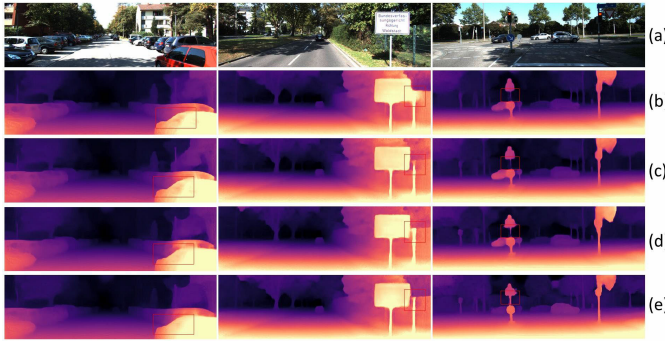


Fig. 5. Qualitative ablation study of the proposed region-sensitive reconstruction loss (RSR) and bidirectional distillation loss (BD) on KITTI dataset. (a) Target images, (b) Baseline, (c) Baseline + [11], (d) Baseline + RSR, (e) Ours.

drift ($^{\circ}/100m$) on length from 100, 200, ..., 800m. The ATE is a measure of global consistency between two trajectories, comparing absolute distances between ground truth and predicted poses at each point in time. On sequence 09, our ATE is 0.007 compared to the baseline of 0.020, while on sequence 10, our ATE is 0.005 compared to the baseline of 0.014. Additionally, since our method separately estimates multiple pose combinations and selects the optimal result from them, there is an improvement in performance on both translation and rotation metrics against other compared methods.

3) *Region-Sensitive Reconstruction Loss*: As stated in Sec. III-D, our region-sensitive reconstruction loss can be regarded as an improved version of feature-level loss based on the work [11]. Although the processing flow is only different in the order of viewpoint synthesis and feature extraction, it has completely different meanings. Reference [11] just leverages the pre-trained encoder as a pure feature extractor, but we excavate the potential capabilities of the model to identify the poorly reconstructed frames as anomaly. Table V verifies our effectiveness. For fair comparison, both methods are introduced into the same baseline (also the first row of Table III) and we achieve better performance. Fig. 5 additionally shows the visual results. Focusing on the comparison between (c) and (d), thanks to the pre-trained model’s holistic semantic

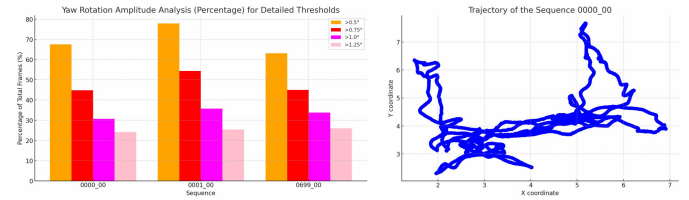


Fig. 6. Statistical results on indoor dataset [57]. We use the ground truth poses to analyze the rotation between consecutive frames. The table specifically shows the percentages of rotation angles exceeding 0.5° , 0.75° , 1° , and 1.25° in sequences 0000, 0001, and 0699 (left), and the overall trajectory plot for sequence 0000 (right).

understanding of objects and sensitivity to anomalies, our method performs better in recovering objects with large color differences (the first column) or fine structures (the second and third columns).

4) *Reconstruction-Guided Bidirectional Distillation Loss*: Table VI shows the performance comparison between teacher and student networks at the independent training stage (training from scratch) and collaborative training stage with bidirectional distillation. At beginning, the performance of the teacher is better than that of the student owing to incorporate more associative data (e.g., stereo images) as input. Through collaborative learning with our proposed reconstruction-guided bidirectional distillation loss, the student and the teacher can learn from each other to further enhance their performance, and ultimately achieve comparable results. Due to slightly different parameter settings or other loss function settings, their performance will inevitably have some fluctuations, resulting in situations similar to those in the table. However, their performances are generally similar. Fig. 5 (e) also demonstrate its effectiveness. With bidirectional distillation, the recovered details are more accurate, such as the smoothness of the car in the first example and the structure of the sticks in last two examples.

C. Adaptability and Generalization

First, we validate the adaptability of our proposed method on NYU v2 dataset, which features challenging rotation-dominant indoor scenes. As shown in Figure 6, unlike

TABLE VII
ADAPTABILITY ON NYU v2 DATASET [50]

Method	Error Metric (lower is better)			Accuracy Metric (higher is better)		
	RMS	Log10	AbsRel	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
Monodepth2 [7]	0.639	0.074	0.176	0.734	0.937	0.983
Zhou et al. [55]	0.712	0.086	0.208	0.674	0.900	0.968
Zhao et al. [56]	0.686	0.079	0.189	0.701	0.912	0.978
Bian et al. (Baseline) [54]	0.608	0.068	0.159	0.772	0.939	0.982
Ours	0.568	0.063	0.149	0.796	0.945	0.985

TABLE VIII
MODEL SIZE, RUNTIME, AND PERFORMANCE COMPARISON OF DIFFERENT METHODS ON MAKE3D DATASET

Method	Size (MB)	Time (ms)	Error Metric (lower is better)			
			Sq Rel	RMSE	RMSE log	Abs Rel
Monodepth2 [7]	57	32	3.378	7.252	0.163	0.321
HR-Depth [42]	56	38	2.944	6.857	0.157	0.305
DIFFNet [28]	42	58	2.901	6.753	0.153	0.298
Lite-Mono-8M [46]	34	46	3.145	7.106	0.158	0.309
Ours (ResNet-18)	57	32	2.976	6.923	0.159	0.301
Ours (HRNet-18)	44	56	2.757	6.669	0.152	0.292



Fig. 7. Qualitative comparison on NYU v2 dataset: (a) Color image, (b) Bian et al. (Baseline) [54], (c) Ours, (d) groundtruth.

outdoor scenes where vehicles mainly move straight and only turn at lane changes or intersections, the pose variations in indoor scenes are more complex, and the opposite situation occurs: the proportion of rotations is much higher than that of straight movements. However, this also represents a form of imbalance between rotation and translation data. Our proposed method is also applicable to this situation. The experiments in Table VII can validate this. We use a recent self-supervised depth estimation network [54] which is also applied to indoor scenes, as our baseline. With our proposed pose-adaptive reconstruction module, we achieve better performance against the Baseline network, and outperform other methods that also test on NYU v2 dataset. Fig. 7 shows the visual comparison. Our method infers more accurate depth values and sharper depth edges than the Baseline.

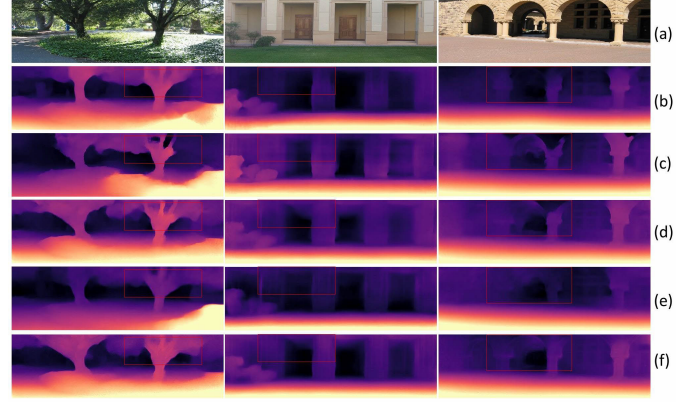


Fig. 8. Qualitative comparison on Make3D dataset: (a) Color image, (b) Monodepth2 [7], (c) HR-Depth [42], (d) DIFFNet [28], (e) Lite-Mono-8M [46], (f) Ours. Regions of red rectangles in predicted depth maps present obvious visual differences among these methods.

Next, we test the cross-dataset generalization ability and apply our model trained on KITTI to Make3D dataset. We compare with other methods among which their models are available and can be tested on Make3D directly. As shown in Table VIII, we achieve the best performance in all metrics, demonstrating the effectiveness of the proposed method in cross-dataset generalization. Fig. 8 shows the qualitative results. In contrast, our method can recover fine structures of objects and provide more reasonable depth values.

D. Runtime

As shown in Table VIII, our core improvement focuses on the loss design and is suitable for assembly on any backbone network, which would not increase the runtime of depth estimation during testing. The running time mainly depends on the size of backbone depth network. Monodepth2 adopts the simplest ResNet-18 as the backbone network and presents the fast runtime (32ms). Utilizing similar backbone, the case ‘Ours (ResNet-18)’ has the same model size and runtime as Monodepth2 but achieves obvious performance improvement. Similar result is concluded on DIFFNet backbone (HRNet-18).

V. CONCLUSION AND FUTURE DIRECTIONS

This paper proposes three novel self-supervised losses via adaptive reconstruction constraints, including pose-adaptive reconstruction loss, region-sensitive reconstruction loss, and bidirectional distillation loss. We achieve superior performance compared with other state-of-the-art methods.

For future directions, we aim to address the failure of photometric loss in adverse conditions such as nighttime. We will investigate the use of alternative loss functions and data augmentation techniques that can enhance the model's performance under low-light conditions.

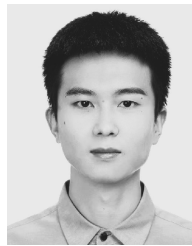
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